Fiscal Year:	FY 2012	**Task Last Updated:**	FY 06/05/2012
PI Name:	Thiessen, David Brian Ph.D.		
Project Title:	Phase Separation and Phase Change Using Helically-Supported Capillary Channels		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pullman	**State:**	WA
Zip Code:	99164-2814	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	05/15/2014
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 5/15/2014 per NSSC information (Ed., 4/5/2013) NOTE: New end date is 5/31/2013 per NSSC (Ed., 4/18/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Marston, Philip (Washington State University)		
Grant/Contract No.:	NNX09AI82G		
Performance Goal No.:			
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Task Description:	Wicking structures with low resistance to axial flow are currently used in heat pipes and may have other applications in microgravity associated with phase separation and phase change technologies. The proposed work involves a fundamental study of a novel capillary flow channel consisting of a helical-wire support that is potentially useful in technologies and benefits from having a symmetry that simplifies the analysis of stability and flow dynamics. The channel geometry makes it particularly easy to measure the capillary dynamics on a global scale using high-speed imaging and image analysis. It is proposed to study the capture of droplets from a flowing air stream by a liquid-filled channel and the capture of bubbles from a flowing liquid stream by a gas-filled channel. Phenomena to be understood include the drop or bubble coalescence event and the subsequent response of the channel as capillary waves propagate from the impact zone dissipating energy. A model for capillary wave dynamics and for dynamic flows in these channels will be developed. Models for drop and bubble capture will also be developed that will be coupled to the channel dynamics model. Studies of phase change at an extended meniscus supported by the helical wire are also proposed. In all cases the basic component to be studied is the capillary channel supported by a helical wire and connected to a constant-pressure reservoir. The behavior of multichannel arrays for phase separation will also be investigated. Parabolic flight experiments are planned and ISS facilities are identified for possible future experiments. The understanding gained from this project is expected to be applicable to other channels that are based on the same fundamental physics. The experimentally validated physics-based models developed from this work might help guide the design of heat pipes and of novel phase-separation devices.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The experimentally validated physics-based models developed from this work might help guide the design of heat pipes and of novel phase-separation devices. The channels being developed have potential as variable-pitch superhydrophobic channels with small resistance to flow.
Task Progress:	The primary objective of this project is to develop a physics-based understanding of drop and bubble capture by helically-supported capillary channels for applications to phase-separation in low gravity. Drop capture by liquid-filled channels is thought to be enhanced by energy transport from the impact zone by capillary waves. A fundamental theory for capillary waves on these channels has now been completed and the predicted capillary wave speeds compare well with measurements. The theory, based on the Bloch theorem, treats a liquid-filled spring as a liquid cylinder with the free surface constrained by an array of equally-spaced rings. As a component of the final theory for the channel, a theory of the scattering of axisymmetric capillary waves on a liquid cylinder by an infinitesimal wire ring that constrains the free surface was developed and this theory can have broader application to the capillary dynamics of minimally contained liquids in low gravity. As an example we have applied the results for single-ring scattering to predict the resonant frequency of liquid bridges using a ray approach leading to simple analytical expressions that agree well with more exact theories in short- and long-wave limits. A highly-automated ground-based experimental apparatus has been completed for measuring drop capture dynamics and wave dynamics for single drops impacting on a single channel. New data is presented from this experiment including drop capture efficiency versus impact parameter and capillary wave speed versus pitch. A measurement of the percent of drop mass absorbed versus impact parameter shows nearly 100% absorption at low impact speed over a range of impact parameter. A method to combine high-speed video images of different sections of the channel into a panorama has allowed us to track capillary wave packets spatially and temporally over a 9-cm section of the channel. This experimental ability will be useful for validating one-dimensional models of channel dynamics in the future. Flow resistance in the channels is an important parameter in designing phase-separation equipment since it limits the maximum flow per channel that can be achieved. The apparatus for measuring pressure losses for steady flow in a channel was upgraded to avoid errors from flow development at the entrance of the channel. Flow resistance will be measured in uncoated springs and springs coated with a superhydrophobic film. It might be expected that a significant reduction in flow resistance could be achieved with the coated springs. In year two a superhydrophobic coating was identified to be used on springs to enable bubble capture in submerged air-filled springs. This year a system has been developed for uniformly coating springs with the superhydrophobic film. A special lathe was built to turn the spring while spraying the sol-gel solution with an air brush to produce uniformly-coated springs. The ability to coat surfaces with a superhydrophobic coating using an air brush was a key enabling technology in the flight experiment described below. In order to advance the technology readiness level (TRL) of phase separation technology based on multi-channel arrays, a parabolic aircraft flight experiment was designed and built to demonstrate drop removal from a flowing gas phase using a 32-channel array. The flight experiment was performed on May 3 and 4, 2012. The data is currently being analyzed but preliminary indications suggest that the basic concept can work. Technical problems with filling the manifolds in the amount of time available on parabolic flight and fundamental problems with air entrapment in the manifolds prevented the filling of all channels. In spite of these problems, the complete filling of some channels was confirmed, and the ability of the array to coalesce all drops in the two-phase flow into large consolidated masses was also confirmed. The failure to fully prime the channels prevented significant removal of water from the two-phase flow.
Bibliography Type:	Description: (Last Updated: 09/14/2018)
Abstracts for Journals and Proceedings	Thiessen D, Zhang L. "Scattering of axisymmetric capillary waves on a liquid cylinder by a pair of surface-pinning rings and the relation to liquid-bridge resonance." 64th Annual Meeting of the APS Division of Fluid Dynamics, Baltimore, MD, November 20-22, 2011. Bulletin of the American Physical Society 2011 Nov;56(18):G15.00009. , Nov-2011
Abstracts for Journals and Proceedings	Zhang L, Thiessen DB. "Propagation of axisymmetric capillary waves along a periodically-pinned liquid cylinder: Bloch waves and scattering." 64th Annual Meeting of the APS Division of Fluid Dynamics, Baltimore, MD, November 20-22, 2011. Bulletin of the American Physical Society 2011 Nov;56(18):G15.00008. , Nov-2011
Articles in Peer-reviewed Journals	Zhang L, Thiessen DB. "Capillary wave scattering from an infinitesimal barrier and dissipation at dynamic contact lines." Journal of Fluid Mechanics. In press as of June 2012. , Jun-2012